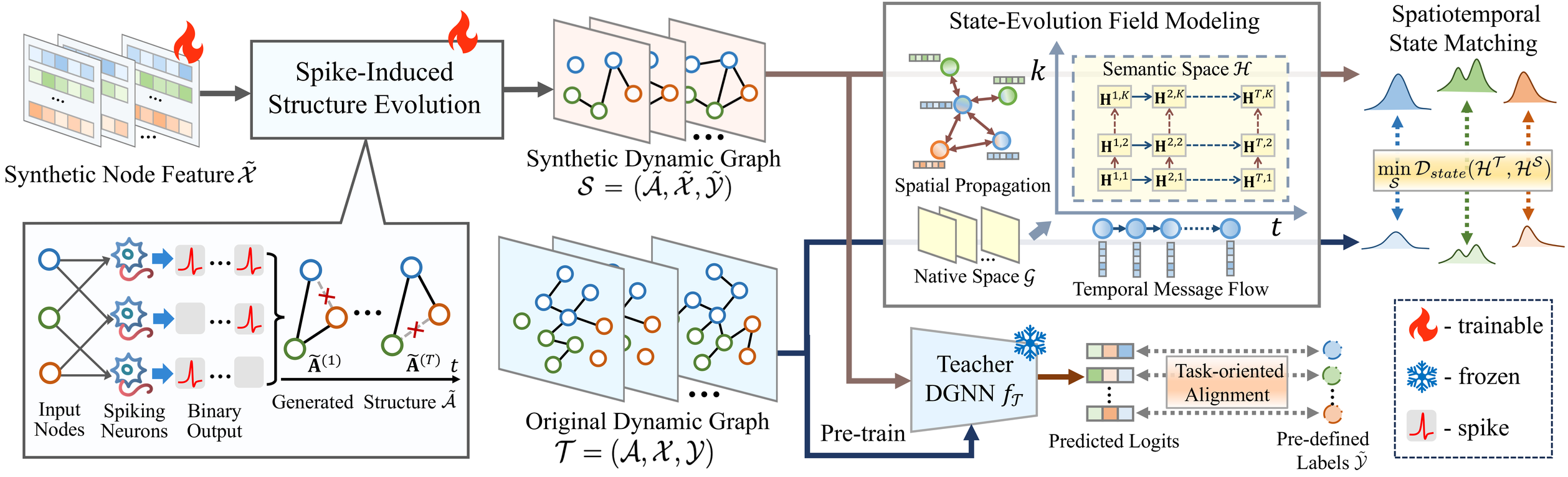
Synthetic Node Feature $\tilde{\mathcal{X}}$
Spike-Induced Structure Evolution
Synthetic Dynamic Graph
$\mathcal{S} = (\tilde{\mathcal{A}}, \tilde{\mathcal{X}}, \tilde{\mathcal{Y}})$
State-Evolution Field Modeling
Spatial Propagation
Semantic Space $\mathcal{H}$
$\mathbf{H}^{1,K}$ $\mathbf{H}^{2,K}$ $\mathbf{H}^{T,K}$
$\mathbf{H}^{1,2}$ $\mathbf{H}^{2,2}$ $\mathbf{H}^{T,2}$
$\mathbf{H}^{1,1}$ $\mathbf{H}^{2,1}$ $\mathbf{H}^{T,1}$
k
t
Native Space $\mathcal{G}$
Temporal Message Flow
Spatiotemporal State Matching
$\min_{\mathcal{S}} \mathcal{D}_{state}(\mathcal{H}^{\mathcal{T}}, \mathcal{H}^{\mathcal{S}})$
Input Nodes
Spiking Neurons
Binary Output
$\tilde{\mathbf{A}}^{(1)}$
$\tilde{\mathbf{A}}^{(T)}$
t
Generated Structure $\tilde{\mathcal{A}}$
Original Dynamic Graph
$\mathcal{T} = (\mathcal{A}, \mathcal{X}, \mathcal{Y})$
Pre-train
Teacher DGNN $f_{\mathcal{T}}$
Task-oriented Alignment
Predicted Logits
Pre-defined Labels $\tilde{\mathcal{Y}}$
- trainable
- frozen
- spike